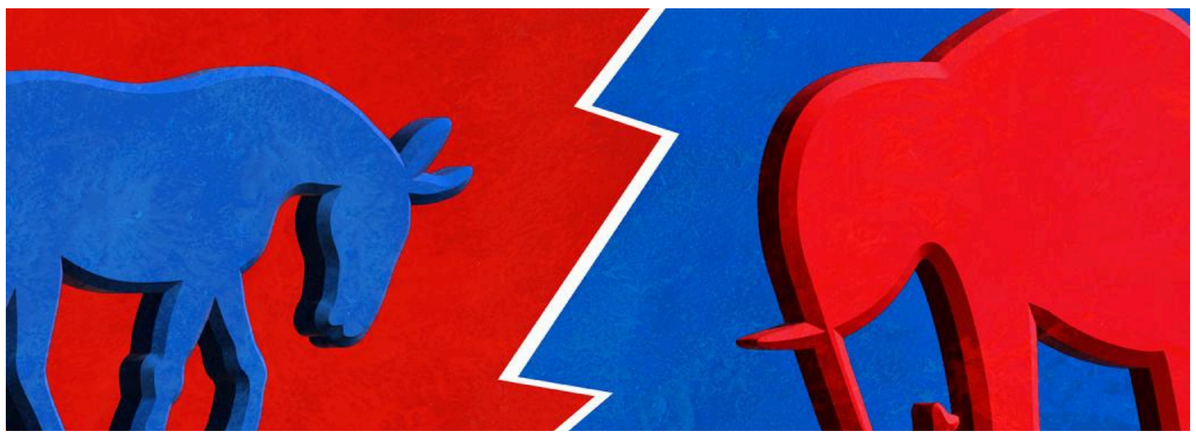
Polarized by Design

A Phase Diagram of Opinion Polarization on Adaptive Hebbian Networks

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Agreement strengthens ties; disagreement erodes them. On a network whose wiring never changes, that single rule is enough to split a society in two.

Forwarder (F) Debunker (D) cross-tie



INTRODUCTION & MOTIVATION

Polarization splits a population into camps that increasingly disagree — and stop talking. Classical opinion models (bounded confidence, cultural dissemination, majority rule) show local imitation can give either con- sensus or division; adaptive-network models (Holme–Newman; Baumann et al.) let agents *rewire* away from disagreement, forming echo chambers.

Our twist. The wiring never changes — no tie is cut or created. Only its *strength* adapts, Hebbian-style. We ask not *whether* polarization appears, but *when*, how *sharply*, and how it depends on *structure*.

POLARIZATION, MADE VISIBLE

We summarise the system by the mean weight of *cross-opinion* (F–D) ties:

$$P(T) = 1 - \bar{w}_{\times}, \quad \bar{w}_{\times} = \frac{1}{|E_{\times}|} \sum_{(i,j)\in E_{\times}} w_{ij}.$$

P=1:every bridge dead(full polarization). $P \approx 0.5$:bridges at baseline—*nothing happened*. Since w0 = 0.5, the value 0.5 means "no dynamics," not "half polarized."

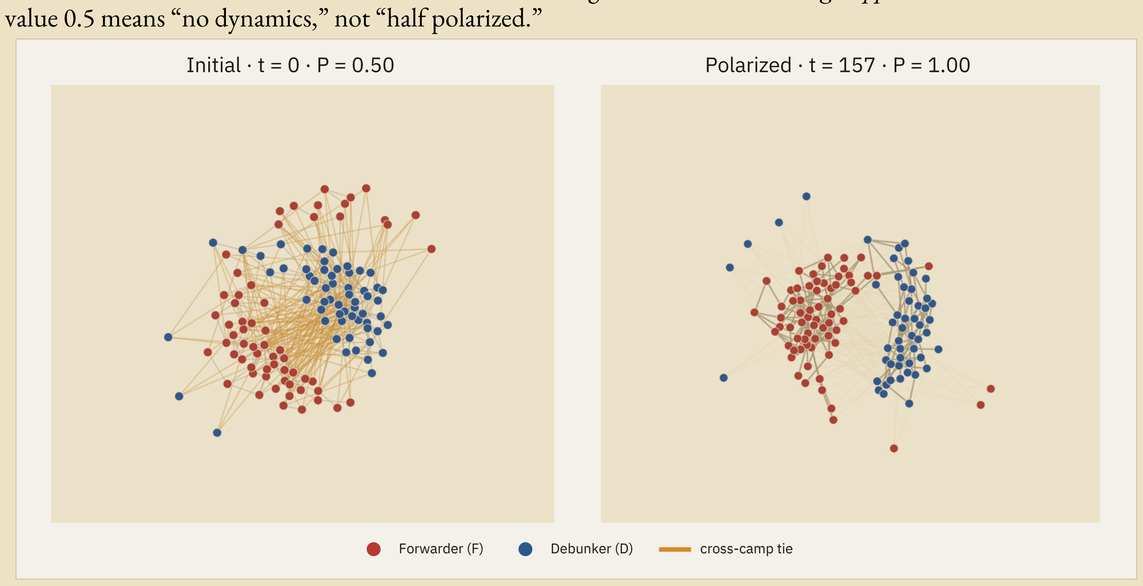


Figure 1. The live model on a Barabási–Albert network (N=120, $\langle k \rangle$=6; S=32, a=0.05). *Left*: the initial 50/50 society (t=0, P =0.5). *Right*: the *same* graph at t=157 (P =1) — every cross-camp tie has decayed to a silent bundle. Edges are neither created nor destroyed; the graph did not break, its bridges went quiet.

THE MODEL & ITS DERIVATION

N agents sit on a fixed graph; each holds opinion F or D. Every edge carries a trust weight $w_{ij} \in [0, 1]$ (all start at w0=0.5). Each agent owns a *stubbornness* $S_i \in [1, 100]$ — its resistance to changing its mind: S_i=1 is minimal resistance (easily swayed), S_i=100 near-immovability. The base model fixes one global S, the population's average.

From one interaction. Take two adjacent agents who disagree. When j tries to persuade i, the attempt is gated by i's *own* stubbornness: it lands when a fresh draw $r \sim U(0, 1)$ clears the trust-to-stubbornness ratio, Sir<w_{ij}. Asr is uniform,

$$p_{ij} = \Pr\left(r < \frac{w_{ij}}{S_i}\right) = \frac{w_{ij}}{S_i}.$$

i

Every node is shielded by its *own* S; the symmetric rule governs i persuading j.

To the neighbourhood. An agent usually faces several dissenters, taken to act *independently*, and a single success flips it. So it holds firm only if *every* attempt fails — the product of $(1 - p_{ij})$:

$$p_{\text{flip}}(i) = 1 - \prod_{j:\, s_j \neq s_i} \left(1 - \frac{w_{ij}}{S_i}\right).$$

To a stochastic model. Every node draws one coin and flips with probability p flip, all synchronously, then each edge adapts by a fixed step a, clipped t

$$w_{ij} \leftarrow \Pi_{[0,1]}\left(w_{ij} + a\,\sigma_{ij}\right), \quad \sigma_{ij} = \begin{cases} +1 & \text{agree} \\ -1 & \text{disagree,} \end{cases}$$

making w=0 (dead) and w=1 (saturated) the absorbing extremes.

PARAMETER INTUITION

S — stubbornness. 1/S is a tie's *persuadability*: at S=1 a full-trust friend converts you almost surely; at S=100, almost never. Low S pushes toward easy consensus; high S freezes opinions in place.

a — reshaping rate. How fast trust moves. Large a snaps weights to a near-binary {0, 1} within a handful of steps; small a leaves them hovering near w_0 for a long time.

A dynamical race. On every cross-cutting tie persuasion and decay compete. If persuasion wins, the bridge converts to agreement before it weakens; if decay wins, the weight reaches 0 and *permanently* insulates the pair — and because a dead tie rarely revives, that is the engine of irreversibility. The winner, set by the ratio of a to S, selects one of two absorbing states: frozen polarized (two cliques) or churning mixed.

EXPERIMENTAL DESIGN

The fate of a society is a tug-of-war between two rates — how fast ties *reshape* (a) versus how hard opinions *resist* (S) — so rather than guess a threshold, we chart the entire plane. The grid is *logarithmic* because the transition spans orders of magnitude; we replicate across three architectures and four degrees to tell *universal* behaviour from a one-topology artifact; and we average many random seeds per cell because, near the transition, a single stochastic run is uninformative. 462,780 simulations in all.

THE PHASE DIAGRAM

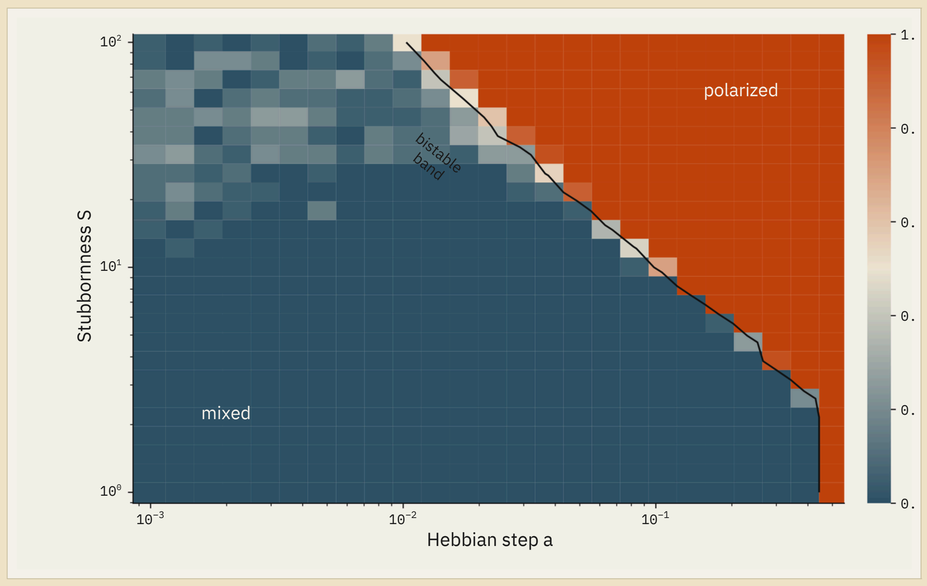


Figure 2. Fraction of societies that polarize over the (a, S) plane (Erdős–Rényi, $\langle k \rangle$=8). A mixed corner, a polarized field, and a diagonal bistable band (the 0.5 contour) between them.

THREE REGIMES — AND A COIN FLIP

Run-to-run variance σ_{runs} separates them: ≈ 0 in both deterministic regimes, large only in the bistable band.

Regime	(a, S)	med. P	frac. pol.	σ_{runs}
Det. polarized	(0.25, 100)	1.00	1.0	0.00
Bistable	(0.009, 48)	0.28	0	0.33
Det. mixed	(0.017, 4.4)	0.34	0.1	0.01

Microstructure. The divided state is total — pooled, *every* cross tie is dead ($\bar{w}_{\times}$=0) and *every* in-group tie saturates ($\bar{w}_{=}$=1). *"Deterministic" = predictable, not polarized*: only in the band is the fate uncertain — two identical societies, opposite destinies.

CONNECTIVITY & ARCHITECTURE

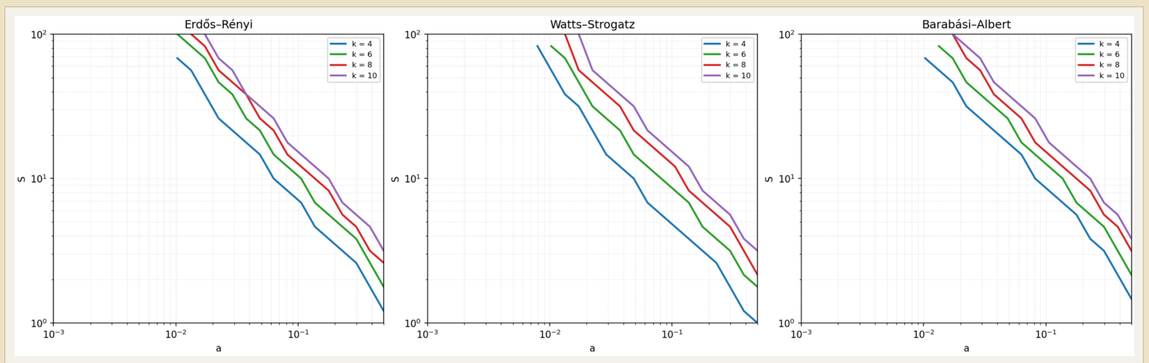


Figure 3. Polarization boundary in (a, S) for each topology and degree: higher $\langle k \rangle$ lifts the boundary (more stubbornness needed to divide).

Connectivity is protective — S∗ rises with ⟨k⟩: more bridges must die before fragmentation. **Architecture matters** — small-world divides most (0.357), random middle (0.325), scale-free resists most (0289); hubs anchor many cross-ties. Given enough stubbornness, division is *inevitable for every architecture*.

S∗(a ≈0.04)	4	6	8	10
Barabási–Albert	14.6	20.5	27.2	33.8
Erdős–Rényi	11.2	16.4	21.6	27.8
Watts–Strogatz	7.5	12.8	18.5	23.8

DISCUSSION

A society here meets one of three fates: deterministically united, deterministically divided, or bistable — poised between, its outcome a coin flip. The microstructure sharpens the contrast: the divided state is *frozen silence* (median 0 opinion changes; all bridges dead, in-group ties saturated), while the united state is *perpetual churn* — the median agent changes its mind on ~2 of every 5 steps, indefinitely.

Connectivity buys resilience. Denser societies polarize *less* easily, all else equal: each extra cross-cutting tie is one more bridge that must be starved to zero before the camps can separate, and an agent ringed by many dissenters feels more competing pulls at once — so the same stubbornness that splits a sparse society leaves a dense one intact. The split is a *threshold* effect, set by a critical stubbornness S∗ above which division becomes inevitable for every architecture we tried.

Influence, not infrastructure. Because the topology never moves, polarization here is purely a story about the *strength* of contact: where a platform fixes who-follows-whom but interaction intensity is free to adapt, echo chambers can form with no rewiring at all. The model's signature prediction is the bistable regime itself — two societies indistinguishable in every parameter diverging to opposite fates through nothing but the accumulated accidents of who changed their mind first.

LIMITATIONS & EXTENSIONS

Limits. Binary opinions; homogeneous ; polarization read as cross-tie death. Weights adapt but *topology does not*. The baseline $P \approx 0.5$ is not "half polarized," and under bistability the *mean* describes no run. Hub mechanism conjectural; N =200 throughout.

Extensions — each unlocks behaviour the present model cannot show:

Heterogeneity. Draw S from a heavy-tailed law: a few near-immovable *zealots* would seed and anchor each camp.

Structure evolution. Let a weight threshold *cut and create* edges — the *topology* itself fragments into true echo chambers.

Asymmetric a. A larger step for one camp — one side radicalizes faster, modelling a lopsided media ecosystem.

Continuous opinions. Beliefs on (−1, 1) with a pull to the median voter — graded moderation replaces all-or-nothing flips.

Also: continuous-time (Gillespie) updating, pair / quenched mean-field, triadic closure, finite-size scaling of S_*.

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